

Customer Churn Prediction API using Flask and Docker

Project Objective

Wrap a trained machine learning model inside a Flask API and containerize it using Docker for deployment.

Technologies Used

Python, Flask, Scikit-learn, Docker, NumPy, Pandas, and Joblib.

Project Features

REST API for prediction
Docker containerization
JSON request and response handling
Model inference endpoint

Project Structure

```
customer-churn-api/  
■■■ app.py  
■■■ model.pkl  
■■■ requirements.txt  
■■■ Dockerfile  
■■■ train_model.py  
■■■ sample_request.json
```

API Endpoint

POST /predict

Example Request

```
{  
  "age": 42,  
  "monthly_charges": 85,  
  "tenure": 24  
}
```

Example Response

```
{  
  "prediction": "Churn"  
}
```

Run Locally

1. Install dependencies using `pip install -r requirements.txt`
2. Train the model using `python train_model.py`
3. Run API using `python app.py`

Docker Commands

Build Image:
`docker build -t churn-api .`

Run Container:
`docker run -p 5000:5000 churn-api`

curl Demo

```
curl -X POST http://localhost:5000/predict -H "Content-Type: application/json" -d  
"{\"age\":42,\"monthly_charges\":85,\"tenure\":24}"
```

Deliverables

Flask API
Dockerfile
requirements.txt
Saved model file
README documentation
Sample request and response

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API Summary Table

Component	Description
Framework	Flask

Containerization	Docker
Endpoint	/predict
Model Type	Random Forest
Input Format	JSON
Output Format	JSON